Bershadsky–Polyakov Koszul self-duality and the hook-type transport

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Abstract

The Bershadsky–Polyakov algebra $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent orbit. The hook partition $(2,1)$ is self-transpose, and the Feigin–Frenkel level involution sends k to $k' = -k-6$. Under the subregular DS/bar-transport hypothesis, this gives $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ away from the critical level. The conductor theorem in the standard Fehily–Kawasetsu–Ridout OPE normalisation is the rational-function identity $K_{\mathcal{B}}(k) \equiv 50$ in $\mathbb{Q}(k)$; equivalently, $c(\mathcal{B}^k) - 25$ is odd in $(k+3)$. The fixed point $k=-3$ of $k \mapsto -k-6$ is the critical pole, so $\kappa(\mathcal{B}^{-3}) = 25/6$ denotes the principal-value symmetric limit. The Bershadsky–Polyakov modular characteristic is $\kappa_{\mathcal{B}} = c_{\mathcal{B}}/6$ in the BRST generator-orbit normalisation used here. The shifted quadratic-numerator invariant $2 - 24(k+1)^2/(k+3)$ gives the auxiliary conductor 196; it is not the FKR central charge unless the shifted stress tensor is specified. The complementarity sum $\kappa_{\mathcal{B}} + \kappa'_{\mathcal{B}} = K_{\mathcal{B}}/6 = 25/3 \neq 0$ is a non-principal instance of a non-vanishing complementarity in the W-algebra landscape. The shadow obstruction tower is class M on the T-line and class G on the J-line.

Remark 0.1 (Polynomial conductor identity). The assertion $K_{\mathcal{B}} = 50$ means $K_{\mathcal{B}}(k) \equiv 50$ as a rational function, not as a level evaluation. The fixed point $k=-3$ is critical; the associated number is the symmetric principal value across the pole. The quadratic-numerator convention gives the parallel shifted constant 196 after changing the conformal vector.

1 Introduction

Hook-type partitions in type A, studied by Fehily [3] and developed by Creutzig–Linshaw–Nakatsuka–Sato [4], are closed under transposition. The associated non-principal W-algebras are $W_k(\mathfrak{sl}_N, f_\lambda)$ for hook-type partitions $\lambda = (N-j, 1^j)$. For these partitions the transpose λ^t is again a hook: $(N-j, 1^j)^t = (j+1, 1^{N-j-1})$, and the hook-transpose duality has the form

$$\big(W_k(\mathfrak{sl}_N, f_\lambda)\big)^! \;\cong\; W_{k^\vee_\lambda}(\mathfrak{sl}_N, f_{\lambda^t})$$

for a specific dual level $k^\vee_\lambda$ determined by the Feigin–Frenkel involution of the Levi subalgebra.

The simplest non-trivial case is $N=3$, $\lambda=(2,1)$. Here $\lambda^t = (2,1) = \lambda$: the partition is self-transpose. The corresponding W-algebra is the Bershadsky–Polyakov algebra $\mathcal{B}^k$, the DS reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent. At levels where the completed DS/bar-transport theorem applies, self-transposition gives $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ with $k' = -k-6$. Thus $\mathcal{B}^k$ is the smallest non-principal test case for hook-transpose transport; its conductor, complementarity defect, and mixed T/J shadow depth are computed below.

2 The Bershadsky–Polyakov algebra

2.1 Definition via DS reduction

Definition 2.1 (Bershadsky–Polyakov algebra). Let $k \in \mathbb{C} \setminus \{-3\}$. The *Bershadsky–Polyakov algebra* $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the BRST cohomology of the quantum DS reduction of $V_k(\mathfrak{sl}_3)$ at the minimal nilpotent element $f \in \mathfrak{sl}_3$ corresponding to the partition $(2, 1)$ of 3.

The Jacobson–Morozov $\mathfrak{sl}_2$-triple (e, h, f) for the minimal nilpotent of $\mathfrak{sl}_3$ determines an $\mathrm{ad}(h)$-grading on $\mathfrak{sl}_3$. The centralizer $\mathfrak{g}^f$ has dimension 4, and the DS reduction produces four strong generators:

Generator	Conformal weight	Field	J-charge
J	1	current	0
G^+	3/2	charged half-integral field	+1
G^-	3/2	charged half-integral field	−1
T	2	stress tensor	0

2.2 Central charge

Proposition 2.2 (BP central charge). *The central charge of $\mathcal{B}^k$ is*

$$c(k) \;=\; -\frac{(2k+3)(3k+1)}{k+3}. \tag{2.1}$$

Proof. This is the Feigin–Semikhatov/Kac–Roan–Wakimoto OPE normalisation used by Fehily–Kawasetsu–Ridout [1, 5, 6]. The formula is the central charge attached to the displayed BP stress tensor. □

Remark 2.3 (Admissible-level check). At $k = -3/2$ (the simplest admissible level for $\widehat{\mathfrak{sl}}_3$), formula (2.1) gives $c(-3/2) = 0$. At $k = -1$ one obtains $c(-1) = 1/2$. These are values in the standard FKR conformal-vector convention.

2.3 OPE data

The OPE of $\mathcal{B}^k$ is the Feigin–Semikhatov Bershadsky–Polyakov normal form [1, (A.3)]. The bosonic sector is:

$$T_{(3)}T = \tfrac{c}{2}, \qquad T_{(1)}T = 2T, \qquad T_{(0)}T = \partial T,$$
$$J_{(1)}J = k_J = \frac{2k+3}{3}, \qquad J_{(0)}J = 0, \qquad T_{(1)}J = J, \qquad T_{(0)}J = \partial J.$$

The conformal action on the odd generators is

$$T_{(1)}G^\pm = \tfrac{3}{2}G^\pm, \qquad T_{(0)}G^\pm = \partial G^\pm, \qquad J_{(0)}G^\pm = \pm G^\pm.$$

The charged sector:

$$G^+_{(2)}G^- = (k+1)(2k+3), \quad G^+_{(1)}G^- = 3(k+1)J, \qquad G^+_{(0)}G^- = 3{:}JJ{:} + \tfrac{3}{2}(k+1)\partial J - (k+3)T,$$
$$G^-_{(2)}G^+ = (k+1)(2k+3), \quad G^-_{(1)}G^+ = -3(k+1)J, \quad G^-_{(0)}G^+ = 3{:}JJ{:} - \tfrac{3}{2}(k+1)\partial J - (k+3)T.$$

All generator self-OPEs among G^+G^+, G^-G^- vanish. The maximal OPE pole order among generators is $p_{\max}(\mathcal{B}^k) = 4$ (from $T_{(3)}T$), hence $k_{\max} = 3$ (the collision depth after $d\log$ absorption on the ordered configuration space $\mathrm{Conf}_2^<(\mathbb{R})$). The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{B}^k})$ lives on the ordered bar $B^{\mathrm{ord}}(\mathcal{B}^k) = T^c(s^{-1}\bar{\mathcal{B}}^k)$; the modular characteristics κ_T and κ_J are Σ_2-coinvariants of the channel-decomposed r-matrix.

3 Koszul self-duality

3.1 The Feigin–Frenkel involution

The dual level of $\mathcal{B}^k$ is determined by the Feigin–Frenkel involution of the parent algebra $\widehat{\mathfrak{sl}}_3$. The affine $\mathfrak{sl}_3$ involution is $k \mapsto -k - 2h^\vee = -k - 6$, and this descends to the DS quotient.

Proposition 3.1 (BP Koszul self-duality). *Assume a comparison theorem identifying the completed chiral bar-cobar Verdier dual with the CLNS convolution dual for the subregular $\mathfrak{sl}_3$ hook. For $k \neq -3$, this additional comparison gives $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ with $k' = -k - 6$.*

Proof. The partition $(2,1)$ is self-transpose: $(2,1)^t = (2,1)$. The CLNS/Fehily hook-transpose theorem gives the convolution dual hook surface. The stated Verdier Koszul conclusion follows only after the assumed DS/bar comparison identifies that surface with the completed chiral bar-cobar dual. For $\mathfrak{sl}_3$ with $\lambda = (2,1)$, the Levi subalgebra is $\mathfrak{g}_0 \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_2$, and the Feigin–Frenkel involution on the affine extension gives $k^\vee_{(2,1)} = -k - 2h^\vee = -k - 6$. □

3.2 The Koszul conductor

Definition 3.2 (Koszul conductor). For a Koszul pair $(A, A^!)$ with central charges c and c', the *Koszul conductor* is $K := c + c'$.

Theorem 3.3 (Bershadsky–Polyakov FKR-conductor identity). *In the rational function field $\mathbb{Q}(k)$, the Koszul-conductor function*

$$K_{\mathcal{B}}(k) \;:=\; c(\mathcal{B}^k) + c(\mathcal{B}^{-k-6}) \;=\; -\frac{(2k+3)(3k+1)}{k+3} - \frac{(-2k-9)(-3k-17)}{-k-3}$$

is identically 50*:*

$$K_{\mathcal{B}}(k) \;\equiv\; 50 \qquad \text{as an element of } \mathbb{Q}(k).$$

Equivalently, $c(\mathcal{B}^k) - 25$ is an odd *function of $(k+3)$: $[c(\mathcal{B}^k) - 25] + [c(\mathcal{B}^{-k-6}) - 25] = 0$ in $\mathbb{Q}(k)$. The fixed point $k = -3$ of the level involution $k \mapsto -k - 6$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3) = -3$, where both summands of $K_{\mathcal{B}}(k)$ are singular. The constancy of $K_{\mathcal{B}}(k)$ on $\mathbb{Q}(k) \setminus \{-3\}$ uniquely extends through the removable singularity; in the Bershadsky–Polyakov modular-characteristic normalisation $\kappa_{\mathcal{B}}(k) = \frac{1}{6}c_{\mathcal{B}}(k)$, the symmetric limit of the central charge across* *the pole is* $\lim_{\varepsilon\to 0}\frac{1}{2}[c_{\mathcal{B}}(-3+\varepsilon)+c_{\mathcal{B}}(-3-\varepsilon)] = 25$, *and therefore* $\kappa(\mathcal{B}^{-3}) = 25/6$ *is the principal-value symmetric limit* $\lim_{\varepsilon\to 0}\frac{1}{2}[\kappa(\mathcal{B}^{-3+\varepsilon})+\kappa(\mathcal{B}^{-3-\varepsilon})]$ *across the pole, not a regular function value. The factor* $1/6$ *in* $\kappa_{\mathcal{B}} = c_{\mathcal{B}}/6$ *is the DS-BRST generator-orbit trace*

$$\varrho_{\mathcal{B}} = 1 - \frac{1}{3/2} - \frac{1}{3/2} + \frac{1}{2} = \frac{1}{6}.$$

The polynomial identity $K_{\mathcal{B}}(k) \equiv 50$ *holds globally in* $\mathbb{Q}(k)$ *even though no single real level realises a self-dual central charge* $c = 25$ *as a regular function value (the real equation* $c(\mathcal{B}^k) = 25$ *has complex roots* $k = -3 \pm 2i$*).*

Proof. Direct polynomial-identity computation in $\mathbb{Q}(k)$. Clear the sign $-k-3 = -(k+3)$:

$$\begin{aligned} K_{\mathcal{B}}(k) &= -\frac{(2k+3)(3k+1)}{k+3} + \frac{(2k+9)(3k+17)}{k+3} \\ &= \frac{(2k+9)(3k+17)-(2k+3)(3k+1)}{k+3} = \frac{50(k+3)}{k+3} = 50. \end{aligned}$$

The $(k+3)$ factor cancels as an identity of rational functions, not as a level evaluation. Hence $K_{\mathcal{B}}(k) - 50 \equiv 0$ in $\mathbb{Q}(k)$. The odd-function reformulation follows from $K_{\mathcal{B}}(k) = 2\cdot 25 + [c(\mathcal{B}^k) - 25] + [c(\mathcal{B}^{-k-6}) - 25] \equiv 50$, which forces the bracketed sum to vanish identically. □

Remark 3.4 (Rational-function conductor). The notation $K_{\mathcal{B}} = 50$ denotes the constant rational function $K_{\mathcal{B}}(k) \equiv 50$ in $\mathbb{Q}(k)$. It is not an evaluation at a particular level, and in particular not an evaluation at the critical fixed point $k = -3$.

Warning 3.5 (Central-charge normalisations). Two central-charge expressions occur in computations around the BP reduction:

$$c_{\mathrm{FKR}}(k) = -\frac{(2k+3)(3k+1)}{k+3}, \qquad c_{\mathrm{shift}}(k) = 2 - \frac{24(k+1)^2}{k+3}.$$

The displayed OPEs in this paper use the Fehily–Kawasetsu–Ridout conformal vector, hence c_{FKR} and conductor 50. The quadratic-numerator expression c_{shift} gives conductor 196 for a shifted conformal normalisation; it must not be cited as the FKR central charge unless that stress tensor is defined.

Remark 3.6 (Two formulas reconciled). The two conductors are both rational-function identities:

$$c_{\mathrm{FKR}}(k) + c_{\mathrm{FKR}}(-k-6) = 50, \qquad c_{\mathrm{shift}}(k) + c_{\mathrm{shift}}(-k-6) = 196.$$

The first belongs to the standard BP OPE written above. The second is an auxiliary shifted-normalisation invariant.

Remark 3.7 (FKR normalisation and the conductor invariant). The FKR normalisation displays the BP reflection polynomial $(2k+3)(3k+1)$. Under $k \mapsto -k-6$ the numerator difference is $50(k+3)$, which is why the conductor is constant.

3.3 Anomaly ratio and complementarity

Definition 3.8 (Anomaly ratio)**.** The *anomaly ratio* of a chiral algebra $\mathcal{A}$ with modular characteristic κ and central charge c is $\rho(\mathcal{A}) := \kappa/c$.

Remark 3.9 ($\kappa_{\mathcal{B}}$, κ_T, and ϱ). The BP calculation involves three distinct quantities.

(a) The full BP modular characteristic is $\kappa_{\mathcal{B}} = c_{\mathcal{B}}/6$. It is the quantity used in Proposition 3.11 and Theorem 3.3.

(b) The Virasoro T-line curvature is $\kappa_T = c_{\mathcal{B}}/2$. It controls the one-dimensional Virasoro shadow tower only.

(c) The residual current line has $\kappa_J = (2k+3)/3$ in the Feigin–Semikhatov normal form.

The signed generator sum ϱ is a BRST generator-orbit trace for (3.1). It is not the Virasoro line curvature.

For the BP algebra, the DS-BRST signed generator-orbit sum has four contributions: J contributes $+1$, the charged G^+ orbit contributes $-\frac{2}{3}$, the charged G^- orbit contributes $-\frac{2}{3}$, and T contributes $+\frac{1}{2}$. These signs are BRST Berezinian signs, not VOA parity assignments. Thus

$$\varrho_{\mathcal{B}} = \frac{1}{1} - \frac{1}{3/2} - \frac{1}{3/2} + \frac{1}{2} = 1 - \tfrac{2}{3} - \tfrac{2}{3} + \tfrac{1}{2} = \frac{6-8+3}{6} = \frac{1}{6}. \tag{3.1}$$

Remark 3.10 (Signed-sum convention for the $G^{\pm}$ orbit). The anomaly-ratio convention used here sums $\varepsilon_i/(2h_i)$ over the charged generator lines in the DS-BRST complex. The fields G^+ and G^- have the same conformal weight and opposite $U(1)$ charges. Their total contribution is $-2/3 - 2/3 = -4/3$.

Proposition 3.11 (Complementarity sum). *For the Bershadsky–Polyakov algebra with the FKR central charge:*

$$\kappa_{\mathcal{B}}(k) + \kappa_{\mathcal{B}}(k') = \frac{c(k)}{6} + \frac{c(k')}{6} = \frac{K_{\mathcal{B}}}{6} = \frac{25}{3}.$$

This is ***nonzero****: the complementarity sum does not vanish. For the Virasoro algebra, $\kappa + \kappa' = 13$; for the BP algebra, $\kappa_{\mathcal{B}} + \kappa'_{\mathcal{B}} = K_{\mathcal{B}}/6$ in the BP modular-characteristic normalisation.*

Proof. Divide the rational-function conductor identity $c(k) + c(k') = K_{\mathcal{B}} = 50$ of Theorem 3.3 by 6. □

Remark 3.12 (Non-principal complementarity). The complementarity sum $\kappa + \kappa' = 0$ holds for affine Kac–Moody algebras (the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ ensures antisymmetry around 0) and for free fields. It fails for all W-algebras with $K \neq 0$: the Virasoro has $K_c = 26$ and $\kappa + \kappa' = 13$, while the BP algebra has $K_c = 50$ and $\kappa_{\mathcal{B}} + \kappa'_{\mathcal{B}} = 25/3$. Thus $\kappa + \kappa' = 0$ is valid for a W-algebra only when its family-specific conductor vanishes.

4 Shadow obstruction tower

4.1 The T-line: class M

The T-line is the one-dimensional primary slice spanned by the Virasoro generator T. On this line, the shadow obstruction tower of $\mathcal{B}^k$ coincides with that of $\mathrm{Vir}_{c(k)}$, because the shadow projections S_r on the T-line are determined entirely by the Virasoro self-OPE.

Theorem 4.1 (T-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the T-line has depth $r_{\max} = \infty$. The tower is class M (mixed). The shadow data on the T-line is:*

$$S_2 = \frac{c}{2}, \qquad S_3 = 2, \qquad S_4 = \frac{10}{c(5c+22)}, \qquad \Delta_T = \frac{40}{5c+22}.$$

The critical discriminant $\Delta_T = 8\kappa_T S_4 = 40/(5c + 22)$ is nonzero on the regular locus $c(5c + 22) \neq 0$, confirming that the shadow metric on the T-line is irreducible there.

Proof. On the T-line, the MC equation projects to the single-line recursion with $\kappa = c/2$, $\alpha = S_3 = 2$, $S_4 = Q^{\text{contact}}_{\text{Vir}} = 10/(c(5c + 22))$ (the Virasoro quartic contact class). The critical discriminant is $\Delta = 8 \cdot (c/2) \cdot 10/(c(5c + 22)) = 40/(5c + 22)$. On the regular locus $c(5c + 22) \neq 0$, the shadow metric $Q_L(t)$ is irreducible and the tower does not terminate. □

4.2 The J-line: class G

The J-line is the primary slice spanned by the $U(1)$ current J. The J-self-OPE is $J_{(1)}J = k_J = (2k + 3)/3$ (the residual U(1) level), and all higher self-OPE modes vanish.

Theorem 4.2 (J-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the J-line has depth $r_{\max} = 2$. The tower is class G (Gaussian). The modular characteristic on the J-line is*

$$\kappa_J = k_J = \frac{2k + 3}{3}$$

where k_J is the residual $U(1)$ level. The cubic and quartic shadows vanish: $S_3^J = S_4^J = 0$.

Proof. The J-self-OPE is purely quadratic: $J_{(1)}J = k_J$ and $J_{(0)}J = 0$ (after $d\log$ absorption, the collision residue is k_J/z). The cubic shadow requires a composite-field contribution from the $J \cdot J$ OPE at mode (0), which vanishes since $J_{(0)}J = 0$. Higher degrees vanish for the same reason: the shadow obstruction tower terminates at degree 2. □

Remark 4.3 (Mixed depth within a single algebra). The Bershadsky–Polyakov algebra has shadow depth ∞ on the T-line and shadow depth 2 on the J-line. This is the first non-principal example of mixed shadow depth within a single algebra. In the principal landscape, each algebra has a single shadow class (Heisenberg is G on every line, $\widehat{\mathfrak{sl}}_N$ is L, Virasoro is M). The BP algebra demonstrates that non-principal W-algebras can exhibit different shadow classes on different generator lines. The global shadow class of $\mathcal{B}^k$ is M (the maximum over all lines), but the per-line classification is $(T : M,\ J : G,\ G^{\pm} : \text{charged})$.

4.3 Sigma-invariant

The sigma-invariant of a Koszul pair measures the "complementarity defect" at each degree:

$$\sigma^{(r)}(k) := S_r^T(c(k)) + S_r^T(c(k')), \qquad k' = -k - 6.$$

For the T-line: $\sigma^{(2)} = c(k)/2 + c(k')/2 = K/2 = 25$. Higher degrees: $\sigma^{(3)} = 2 + 2 = 4$, $\sigma^{(4)} = 10/(c(5c + 22)) + 10/(c'(5c' + 22))$ where $c' = K - c$. These vanish only on the proper algebraic exceptional locus of the single-line recursion.

Proposition 4.4 (Finite-cutoff sigma non-vanishing). *For the Bershadsky–Polyakov algebra, $\sigma^{(2)} \neq 0$ and $\sigma^{(3)} \neq 0$. For each fixed cutoff $R \geq 4$, there is a proper algebraic exceptional locus E_R such that $\sigma^{(r)} \neq 0$ for $2 \leq r \leq R$ on its complement. Uniform non-vanishing for all r is a separate Virasoro-recursion proof obligation.*

Proof. At degree 2: $\sigma^{(2)} = K/2 = 25 \neq 0$. At degree 3: $\sigma^{(3)} = 4 \neq 0$. At degree 4: the quartic sigma-invariant is a nontrivial rational function of k. For a fixed cutoff R, the finite product of the numerator polynomials for $4 \leq r \leq R$ cuts out the proper exceptional locus E_R. □

5 The hook-type transport

5.1 The hook-type surface

The hook-type surface in type A consists of partitions $\lambda = (N-j, 1^j)$ of N. The transpose is $(N-j, 1^j)^t = (j+1, 1^{N-j-1})$, another hook. This surface is closed under transposition.

Definition 5.1 (Self-transpose hooks)**.** A hook partition $(N-j, 1^j)$ of N is *self-transpose* if and only if $(N-j, 1^j) = (j+1, 1^{N-j-1})$, equivalently $N-j = j+1$, i.e., $j = (N-1)/2$. This requires N odd.

For $N = 3$: $j = 1, \lambda = (2, 1)$ (the BP algebra). For $N = 5$: $j = 2, \lambda = (3, 1, 1)$ (a hook-type W-algebra of $\mathfrak{sl}_5$). For $N = 2m+1$: $j = m, \lambda = (m+1, 1^m)$.

Theorem 5.2 (Hook self-duality)**.** *Assume the completed hook-type DS/bar transport theorem for the self-transpose hook $\lambda = (m+1, 1^m)$ in $\mathfrak{sl}_{2m+1}$, including the finite-type Verdier model and the compatibility of the Drinfeld–Sokolov reduction with the bar comparison. Then the W-algebra $W_k(\mathfrak{sl}_N, f_\lambda)$ is Koszul self-dual:*

$$\big(W_k(\mathfrak{sl}_N, f_\lambda)\big)^! \;\cong\; W_{k'}(\mathfrak{sl}_N, f_\lambda)$$

with $k' = -k-2N$, matching the Feigin–Frenkel involution $k \mapsto -k-2h^\vee$ for $\mathfrak{sl}_N$ where $h^\vee(\mathfrak{sl}_N) = N$. The corresponding hook conductor is a separate proof obligation: one must compute the KRW central charge for $\lambda = (m+1, 1^m)$ and prove that $c_\lambda(k) + c_\lambda(-k-2N)$ is constant.

Proof. Self-transposition $\lambda = \lambda^t$ is immediate from Definition 5.1. The transport-to-transpose theorem of [4] then gives the Koszul duality, with the dual level determined by the Feigin–Frenkel involution $k \mapsto -k-2h^\vee$; for $\mathfrak{sl}_N$ the dual Coxeter number is $h^\vee = N$, so $k' = -k-2N$, consistent with Warning 5.3. For the Bershadsky–Polyakov case $N = 3$: $k' = -k-6$, the familiar $\mathfrak{sl}_3$ duality. □

Warning 5.3 (Dual Coxeter number)*.* $h^\vee(\mathfrak{sl}_N) = N$. The Feigin–Frenkel involution is $k \mapsto -k-2h^\vee = -k-2N$. For $\mathfrak{sl}_3$: $k' = -k-6$. Here N is the defining dimension of $\mathfrak{sl}_N$, not the rank; the rank is $N-1$, while $h^\vee(\mathfrak{sl}_N) = N$ for type A_{N-1}.

5.2 The BP algebra as seed

The Bershadsky–Polyakov algebra is the smallest non-principal case where the PBW strong-generation surface can be tested against the DS/bar comparison. Its chiral Koszulity is used here only under the completed BP bar-cobar comparison hypothesis.

Computation 5.4 (Koszul invariants of $\mathcal{B}^k$)**.** The invariants are:

Invariant	Formula	Value
Central charge $c(k)$	$-(2k+3)(3k+1)/(k+3)$	rational in k
Dual level k'	$-k-6$	involution
Dual charge $c(k')$	$50 - c(k)$	by $K = 50$
Koszul conductor K	$c + c'$	50
$\kappa_T(k)$	$c(k)/2$	Virasoro sector
$\kappa_J(k)$	$(2k+3)/3$	$U(1)$ sector
Shadow depth (T-line)	∞	class M
Shadow depth (J-line)	2	class G
$p_{\max}$	4	from $T_{(3)}T$
$k_{\max}$	3	$= p_{\max} - 1$

The entries follow from the central-charge formula, the Feigin–Frenkel involution, and the T- and J-line OPE computations above.

5.3 Comparison with Virasoro and W_3

The BP algebra sits between the Virasoro algebra and the principal W_3 algebra in the DS hierarchy of $\mathfrak{sl}_3$:

$$\widehat{\mathfrak{sl}}_3 \xrightarrow{\mathrm{DS}_{(2,1)}} \mathcal{B}^k \xrightarrow{\text{further reduction}} \mathrm{Vir}_{c(k)}.$$

The principal DS reduction of $\widehat{\mathfrak{sl}}_3$ produces W_3 (which has two generators T, W of weights 2, 3). The minimal DS reduction produces $\mathcal{B}^k$ (four generators of weights 1, 3/2, 3/2, 2). The Koszul conductors are:

$$K_{\mathrm{Vir}} = 26, \qquad K_{\mathcal{B}} = 50, \qquad K_{W_3} = 2(N-1)(2N^2+2N+1)\big|_{N=3} = 100.$$

The shadow class on the T-line is M for all three algebras. The distinguishing feature of $\mathcal{B}^k$ is the J-line, which is class G: the $U(1)$ current has only quadratic shadow data, and the tower terminates immediately. Neither the Virasoro (no J-line) nor W_3 (no J-line) exhibits this phenomenon.

5.4 Self-dual point

Proposition 5.5 (Critical fixed point of the level involution). *The level involution $k \mapsto -k-6$ has fixed point $k = -3$, the critical excluded level. No regular Bershadsky–Polyakov algebra at that level realizes self-duality. The expression $c(-3)$ is singular, so $\kappa(\mathcal{B}^{-3}) = 25/6$ denotes the symmetric principal value across the pole. No real regular level has $c(\mathcal{B}^k) = 25$; the equation has roots $k = -3 \pm 2i$.*

Proof. $c(k) = c(k')$ requires $c = K - c$, hence $c = K/2 = 25$. Solving $-(2k+3)(3k+1)/(k+3) = 25$ reduces to $k^2 + 6k + 13 = 0$, whose discriminant is $36 - 52 = -16 < 0$. The roots $k = -3 \pm 2i$ are complex, so no real regular level realises the self-dual central charge. The formal self-dual point $k = k' = -3$ is the critical level, where the DS reduction degenerates. ☐

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